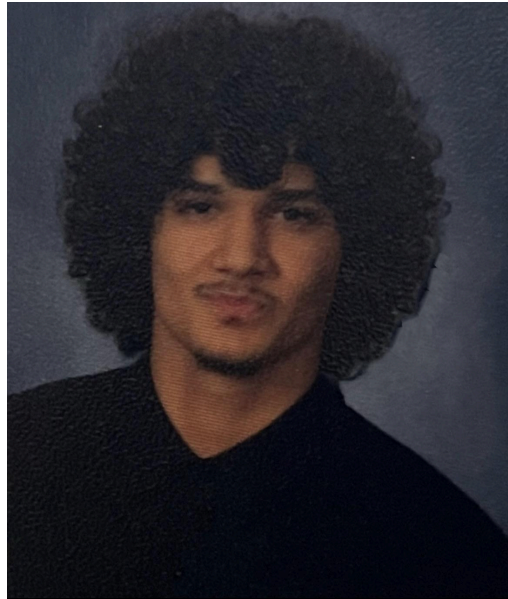


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Rudy's Exploration to a Bionic Arm First Exploration



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Introduction

Hello, my name is Rudy Ramirez. I am a student at Phoenix Coding Academy and an upcoming degree seeking mechanical engineering student at ASU. I am using this opportunity to get a feel for electrical components and what is possible with open source material. What I am trying to get from this project is electrical, software, 3D modeling, and component integration skills.

This project is my first hands-on exploration into robotics and bionics. Starting from zero knowledge of servo motors and mechanical actuation, I worked through research, design, and physical assembly to produce a functioning robotic hand controlled by an Arduino Mega. Every part of the process — from understanding how a real hand moves, to wiring electronics, to printing and assembling the structure — was a learning opportunity I took seriously.

The goal was never just to build something that looked like a hand. The goal was to

understand why it works the way it does, and to build a foundation I can grow from as I move into mechanical engineering at ASU.

Methodology

Understanding How a Hand Works

Before touching any hardware, I needed to understand the anatomy of a real hand — specifically what makes fingers move and what keeps them in position. I studied the mechanics of finger flexion and extension, focusing on how tendons pull the fingers closed and how the opposing tendons on the back of the hand return them to the open position.

From this research I understood that a tendon-driven design — where fishing wire acts as the flexor tendon and elastic cord on the back acts as the extensor — would be the most practical approach for a 3D printed hand at this scale.

Videos Referenced:

[How Do Fingers Work? — Anatomy of the Hand and Tendons](#)
[How Robotic Hands Mimic Human Movement](#)

Articles Referenced:

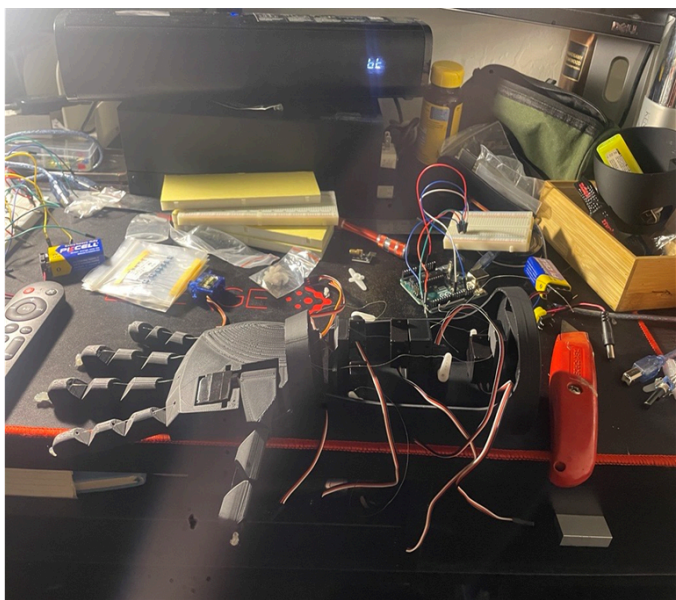
[Tendon-Driven Robotic Hands — ResearchGate](#)
[The Anatomy of Hand Movement — Khan Academy](#)

3D Modeling and Printing

With the mechanical concept understood, I moved into 3D modeling the hand structure. The design needed to house four servo motors in the palm, route fishing wire cleanly through each finger channel, allow full range of motion without binding, and have anchor points on the back for the elastic return tendons.

The parts were printed and assembled before any electronics were attached, allowing me to verify that the finger motion was smooth and that the wire routing didn't bind at any joint.





Electronics and Wiring

Once the structure was printed and assembled, I wired the four micro servos to an Arduino Mega through a breadboard. An external 5V power supply powers the servos directly — running them off the Arduino's onboard 5V pin would damage the board under load. A shared ground between the supply and the Arduino is critical for the signal wires to work correctly.

A button on pin 13 triggers the hand to close. Pressing it rotates all four servos to 90 degrees, pulling the fishing wire and curling the fingers. Releasing the button lets the back tendons return the fingers to the open position.

Servo 1	3
Servo 2	7
Servo 3	8

Servo 4	10
Button	13

Software

The code uses Arduino's built-in Servo library. The button uses INPUT_PULLUP so no external resistor is needed — the pin reads LOW when pressed, triggering all four servos to close simultaneously.

```
#include <Servo.h>

Servo s1, s2, s3, s4;
const int BUTTON_PIN = 13;

void setup() {
  s1.attach(3); s2.attach(7);
  s3.attach(8); s4.attach(10);
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  s1.write(0); s2.write(0); s3.write(0); s4.write(0);
}

void loop() {
  if (digitalRead(BUTTON_PIN) == LOW) {
    s1.write(90); s2.write(90); s3.write(90); s4.write(90);
  } else {
    s1.write(0); s2.write(0); s3.write(0); s4.write(0);
  }
}
```

Reflection

This project gave me direct experience in four areas I set out to develop: electrical wiring, software (Arduino C++), 3D modeling, and integrating all three into a working physical system. The biggest lesson was how much the mechanical design and the electronics depend on each other — a servo cannot do its job if the wire routing binds, and the code cannot help if the ground is not shared correctly.

This is the first exploration. The next step is expanding toward a full bionic arm with more degrees of freedom, individual finger control, and eventually sensor feedback.

